

2020 AMC 10A Problem 17 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 17. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 17

Problem:

Define

$$P(x) = (x - 1^2)(x - 2^2) \cdots (x - 100^2)$$

How many integers n are there such that $P(n) \leq 0$?

Answer Choices:

- A. 4900
- B. 4950
- C. 5000
- D. 5050
- E. 5100

Solution:

We perform casework on $P(n) \leq 0$:

1. $P(n) = 0$ In this case, there are 100 such integers n

$$1^2, 2^2, 3^2, \dots, 100^2$$

	Interval of x	# of Negative Factors	Valid?
	$(-\infty, 1^2)$	100	
	$(1^2, 2^2)'$	99	✓
	$(2^2, 3^2)$	98	
	$(3^2, 4^2)$	97	✓
	$(4^2, 5^2)$	96	
	$(5^2, 6^2)$	95	✓
	$(6^2, 7^2)$	94	
	$(99^2, 100^2)$	1	j
	$(100^2 \cdot \infty)$	0	

2. $P(n) < 0$ There are 100 factors in $P(x)$, and we need an odd number of them to be negative. We construct

Note that there are 50 valid intervals of x . We count the integers in these intervals:

$$\begin{aligned}
 & (2^2 - 1^2 - 1) + (4^2 - 3^2 - 1) + (6^2 - 5^2 - 1) + \cdots + (100^2 - 99^2 - 1) \\
 &= \underbrace{(2^2 - 1^2)}_{(2+1)(2-1)} + \underbrace{(4^2 - 3^2)}_{(4+3)(4-3)} + \underbrace{(6^2 - 5^2)}_{(6+5)(6-5)} + \cdots + \underbrace{(100^2 - 99^2)}_{(100+99)(100-99)} - 50 \\
 &= \underbrace{(2+1) + (4+3) + (6+5) + \cdots + (100+99)}_{1+2+3+4+5+6+\cdots+99+100} - 50 \\
 &= \frac{101(100)}{2} - 50 \\
 &= 5000
 \end{aligned}$$

In this case, there are 5000 such integers n .

Together, the answer is $100 + 5000 = \boxed{(E) 5100}$.

OR

Notice that $P(x)$ is nonpositive when x is between 100^2 and 99^2 , 98^2 and 97^2 , $\dots$, 2^2 and 1^2 (inclusive), because there are an odd number of negatives, which means that the number of values equals

$$((100 + 99)(100 - 99) + 1) + ((98 + 97)(98 - 97) + 1) + \dots + ((2 + 1)(2 - 1) + 1)$$

This reduces to

$$200 + 196 + 192 + \dots + 4 = 4(1 + 2 + \dots + 50) = 4 \cdot \frac{50 \cdot 51}{2} = \boxed{(E) 5100}$$

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